

EMBEDDED COMPUTER · AMD RYZEN EMBEDDED R2312

POE-POWERED SINTRONES SBOX-2624P EMBEDDED COMPUTER

A 150 mm fanless x86 box with a four-port PoE switch built in. Four powered sensors, two 4K displays, four serial ports, and EN 50155 for rolling stock — on 9 to 36 volts.



■ WHY THIS COMPUTER

- **AMD Ryzen™ Embedded R2312**, 2C / 4T with built-in Radeon™ graphics — full x86, so existing Windows and Linux software runs unmodified
- **Integrated 4-port GbE PoE switch** — 15 W per port, 60 W budget; four cameras or sensors on one cable each, no external injector
- **1 × 2.5GbE uplink** alongside the PoE ports, keeping sensor traffic off the network trunk
- **2 × HDMI® at 4K 60 Hz** (HDMI 2.0b) — two independent displays driven from a 150 × 150 × 65 mm chassis
- **4 × serial and 8 × DIO** — 2 × RS-232/422/485 plus 2 × RS-232, 4 × DI and 4 × DO for legacy plant and vehicle hardware
- **Up to 32 GB DDR4 SO-DIMM with ECC support**, TPM 2.0, and a BIOS-configurable hardware watchdog
- **M.2 expansion on three keys** — NVMe or SATA storage, 5G/LTE with dual SIM, and Wi-Fi 6 with Bluetooth 5.2
- **EN 50155 certified**, with CE, FCC Class A and UKCA — fanless from -30 °C to 60 °C on a protected 9–36 V input

4 × GbE

POE PORTS, 60 W

2 × 4K

HDMI 2.0B DISPLAYS

EN 50155

RAIL CERTIFIED

9–36 V

DC INPUT RANGE

■ APPLICATION SCENARIOS



POE SENSOR HUBS

Four powered devices from a single box



INDUSTRIAL AUTOMATION

Fanless control and machine-vision hosts



RAIL & TRANSIT

EN 50155 certified onboard computing



SIGNAGE & HMI

Two independent 4K displays

SINTRONES SBOX-2624P — SPECIFICATIONS

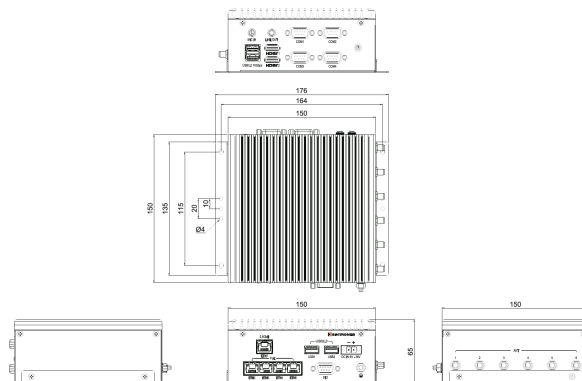


PARAMETER	TECHNICAL DATA
Processor	AMD Ryzen™ Embedded R2312, 2C / 4T
Graphics	CPU built-in AMD Radeon™ graphics
Memory	1 × SO-DIMM DDR4-2400, up to 32 GB (ECC supported with ECC DRAM)
Security	TPM 2.0
Watchdog	System reset, 1–255 sec. or min., configurable in BIOS
Storage	1 × M.2 2280 Key M supporting NVMe or SATA SSD
Video	2 × HDMI®, 4K @ 60 Hz as specified in HDMI 2.0b
Audio	1 × line-out, 1 × mic-in
Ethernet	1 × 2.5GbE, 4 × GbE from Marvell Ethernet switch with PoE, via RJ-45
PoE	PSE Class 3, 15 W per port; 60 W power budget GbE ports share 1GbE bandwidth
USB	2 × USB 3.2 Gen 2 (10 Gbps), 2 × USB 2.0
Serial port	2 × RS-232/422/485, 2 × RS-232
DIO	4 × DI (9–36 V DC), 4 × DO (5 V, 100 mA)

PARAMETER	TECHNICAL DATA
Internal expansion	1 × M.2 3042/52 Key B (USB 3.2, WWAN, dual SIM) 1 × M.2 2230 Key E (PCIe + USB 2.0, Wi-Fi / BT)
Power input	DC-IN 9–36 V (nominal 12 V or 24 V)
Power protection	Over-voltage and reverse polarity protection
Operating system	Windows 10 IoT Enterprise 64-bit LTSC, Windows 11 IoT Enterprise, Ubuntu 20.04 LTS 64-bit, Ubuntu 22.04 LTS 64-bit
Operating temperature	–30 °C to 60 °C with SSD, ambient with 0.6 m/s airflow (barebone system)
Storage temperature	–40 °C to 80 °C
Relative humidity	10% to 90% @ 40 °C, non-condensing
Certification / standard	CE, FCC Class A, UKCA, EN 50155
Construction	Aluminium alloy
Mounting method	Wall mounting
Weight	1,500 g (barebone)
Dimensions (L × W × H)	150 (L) × 150 (W) × 65 (H) mm
Country of origin	Made in Taiwan

OUTLINE AND MOUNTING

SBOX-2624P



UNIT: MM · DRAWINGS NOT TO SCALE

Ordering. SBOX-2624P — AMD Ryzen™ Embedded R2312 with 2 × HDMI® / 1 × 2.5GbE, 4 × GbE PoE / 4 × COM / DC-IN 9–36 V embedded computer. A non-PoE sibling, the SBOX-2624, offers 1 × 2.5GbE + 1 × GbE, 6 × COM, a 2.5" SATA SSD bay and an M.2 CAN bus option. Optional: DDR4 SO-DIMM 8–32 GB (–40 to 85 °C, ECC available); M.2 2280 Key M NVMe 512 GB–2 TB or SATA 128 GB–2 TB; M.2 2230 Key A+E Wi-Fi 6 + BT 5.2; M.2 3042/52 5G/LTE modem; M.2 2242 GPS module.

Notes. The four GbE PoE ports share 1GbE of switch bandwidth; size aggregate sensor throughput accordingly. The 60 W PoE budget is shared across all four ports — verify the combined draw of the connected devices. Operating temperature is specified for the barebone system with an SSD fitted and 0.6 m/s ambient airflow; installed M.2 modules and higher sustained CPU load reduce headroom.

Installation. Mount on a flat surface with the fin stack unobstructed and the specified ambient airflow available; these are fanless chassis and shed heat through the enclosure. Keep the connector face clear for service access. Confirm the DC supply stays within the rated input window under vehicle crank and load-dump conditions before commissioning.

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